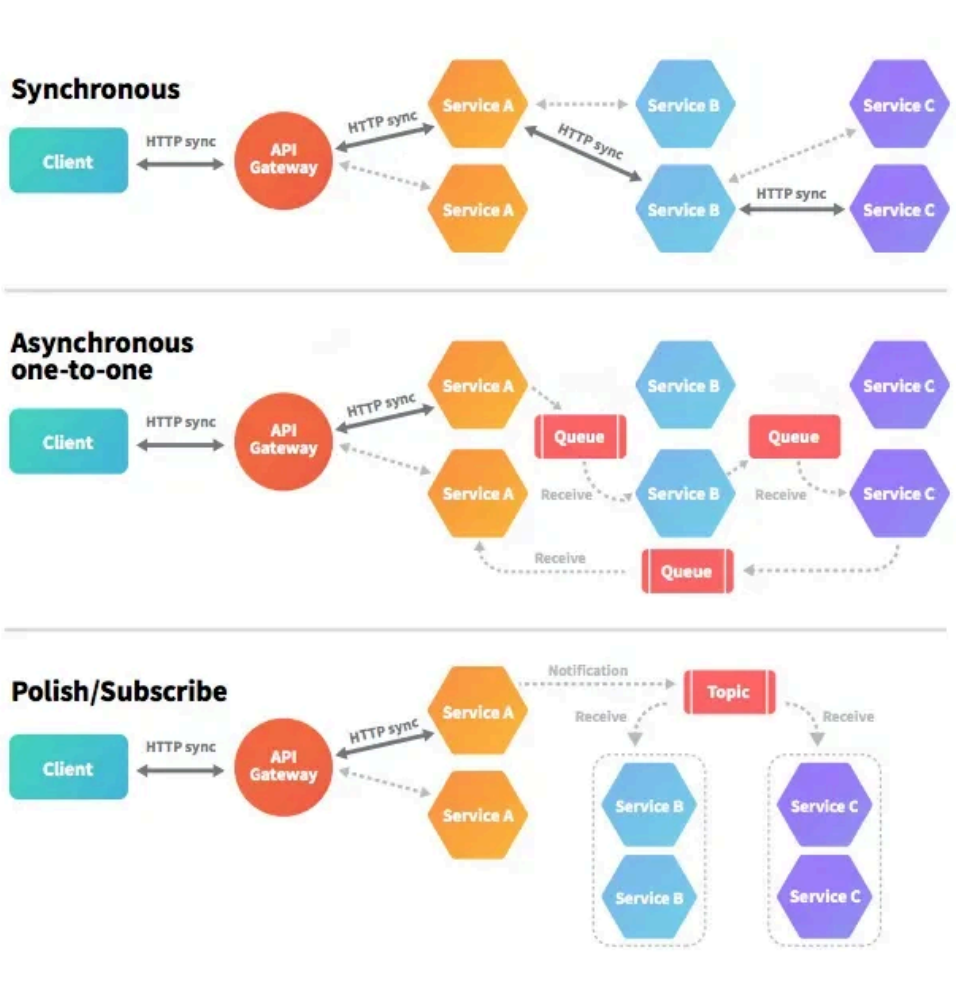
# Asynchronous communications

Asynchronous communication is a key architectural pattern in distributed systems. Unlike synchronous communication, where one service sends a request and waits for a response, **asynchronous communication allows services to continue processing other tasks while awaiting a response**.


Synchronous
Client
HTTP sync
API Gateway
HTTP sync
Service A
Service A
HTTP sync
Service B
Service B
Service C
HTTP sync
Service C
Asynchronous one-to-one
Client
HTTP sync
API Gateway
HTTP sync
Service A
Service A
Queue
Receive
Service B
Service B
Queue
Receive
Service C
Service C
Receive
Queue
Polish/Subscribe
Client
HTTP sync
API Gateway
HTTP sync
Service A
Service A
Notification
Topic
Receive
Receive
Service B
Service B
Service C
Service C

## Benefits of Asynchronous Communication

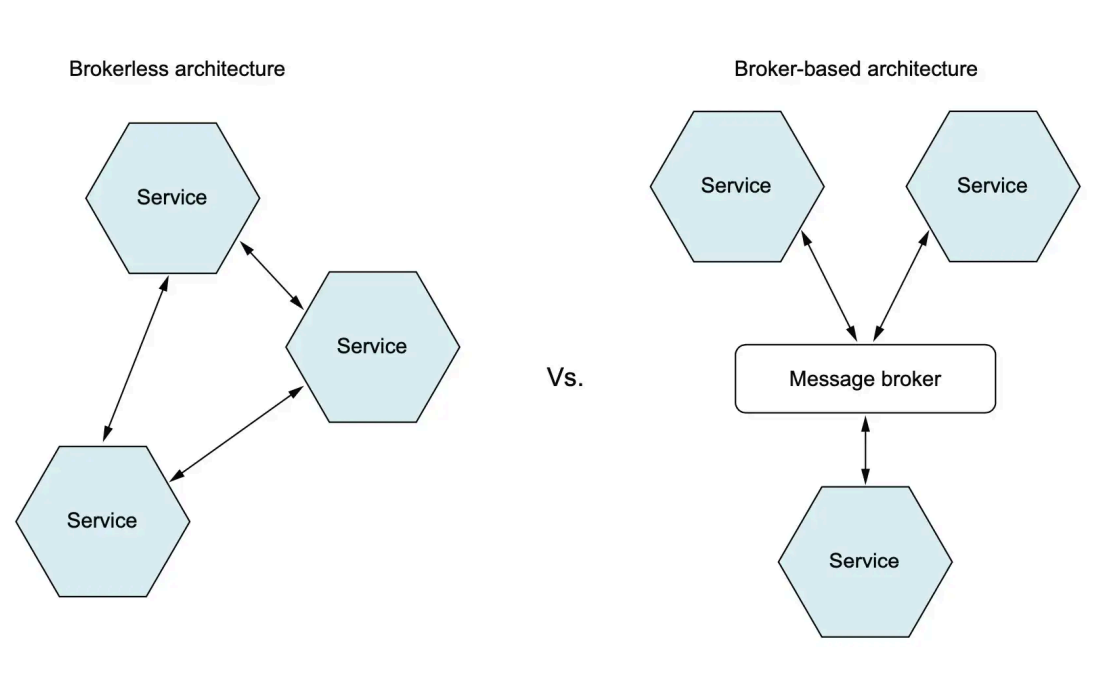
1. **Decoupled Services** – Sender and receiver operate independently, **eliminating temporal coupling**.
2. **Better Resource Utilization** – Threads and memory aren't blocked waiting for responses, **preventing resource exhaustion**.
3. **Improved Scalability** – Services handle tasks independently; **queues distribute load**, allowing horizontal scaling.
4. **Increased Resilience** – Brokers **store and forward messages**, enabling fault-tolerance when services are temporarily unavailable.

## Challenges of Asynchronous Communication

1. **Complexity** – Requires extra infrastructure to **ensure message delivery, processing, and acknowledgment**.
2. **Consistency** – Eventual consistency can lead to **temporary data discrepancies** between services.
3. **Message Ordering** – Maintaining the **correct processing order** is challenging in distributed systems.
4. **Error Handling** – Failures are harder to detect; **retries, dead-letter queues, and monitoring** are needed for reliability.

## Messaging Systems Architectures

There are two primary approaches to asynchronous messaging passing: **broker-based** and **broker-less** (also known as peer-to-peer or direct messaging) systems.


Brokerless architecture
Service
Service
Service
Vs.
Broker-based architecture
Service
Service
Message broker
Service

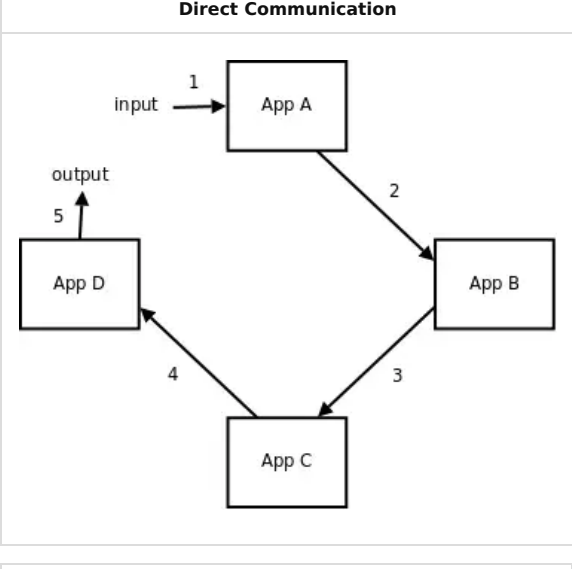
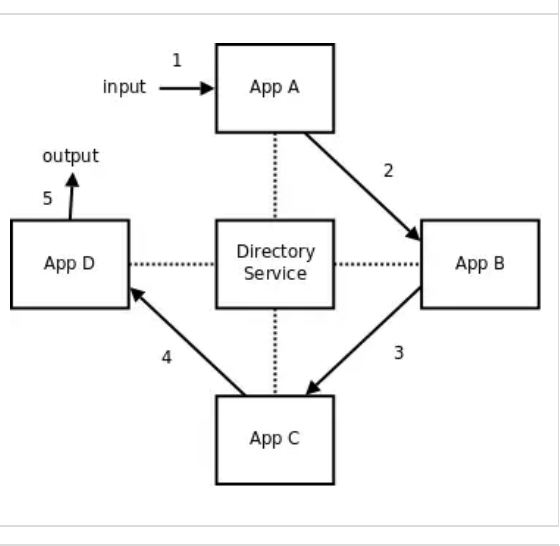
## Brokerless Messaging Systems

**Brokerless messaging** allows applications to communicate directly without a central broker, reducing latency and complexity.

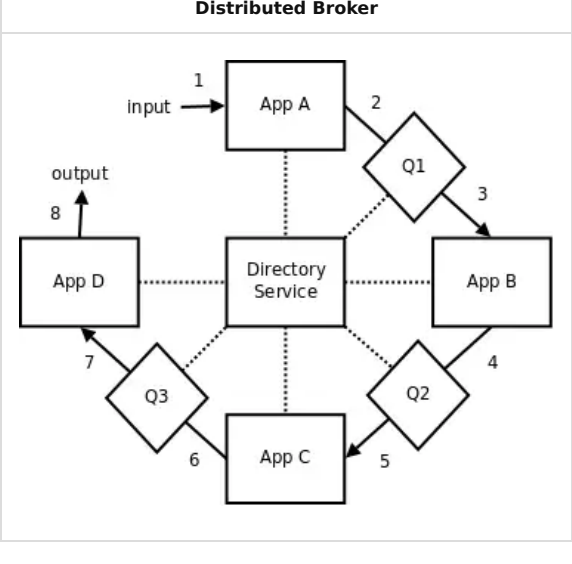
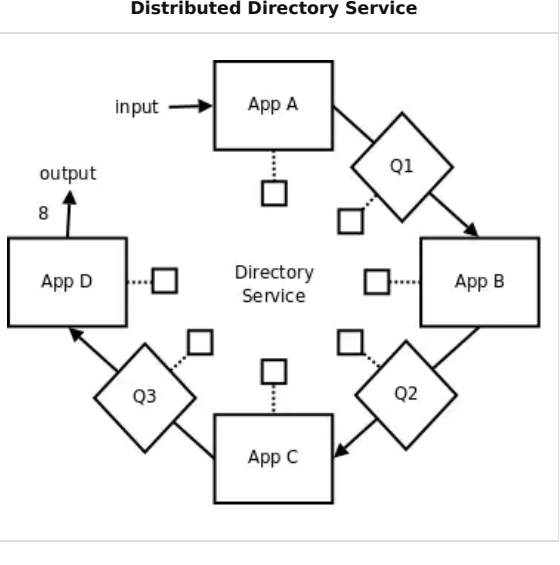
- ZeroMQ: a high-performance library offering asynchronous sockets and patterns like pub/sub, request/reply, and push/pull. It enables fast, low-latency communication across threads, processes, and machines while abstracting networking details.
- NanoMsg: a simpler, modular alternative inspired by ZeroMQ, supporting similar messaging patterns and transports (TCP, IPC, in-process), with a focus on robustness, portability, and maintainability for scalable systems.

**Advantages**

1. **No Single Point of Failure**: Brokerless systems avoid the broker becoming a single point of failure, making the architecture more resilient to certain types of failures.
2. **Low Latency**: Messages travel directly between services, reducing the additional overhead introduced by a broker. This leads to faster communication and lower latency.
3. **More Control**: Direct communication gives services full control over how messages are handled, improving flexibility in handling specific scenarios like retries or error handling.

| Direct Communication | Directory Service |
|---|---|
|  |  |

input
1
App A
2
App B
3
App C
4
App D
5
output
input
1
App A
2
App B
3
App C
4
App D
5
output
Directory Service

| Distributed Broker | Distributed Directory Service |
|---|---|
|  |  |

input
1
App A
2
Q1
3
App B
4
Q2
5
App C
6
Q3
7
App D
8
output
Directory Service
input
App A
Q1
App B
Q2
App C
Q3
App D
8
output
Directory Service

- **Pure Brokerless:** Direct messaging, full control, manual management.
- **Broker as Directory Service:** Broker tracks locations, messages go direct.
- **Distributed Broker:** Lightweight queues handle messages, avoid bottlenecks.
- **Distributed Directory Service:** Replicated directory, no single point of failure, dynamic network.

**Disadvantages**

1. **Tight Coupling** – Services must know how to communicate directly, making changes harder and increasing dependencies.
2. **No Built-in Reliability** – Developers must implement **persistence, retries, and delivery guarantees**, adding complexity.
3. **No Built-in Scaling** – High-throughput scenarios require **custom load balancing and failover mechanisms**.
4. **No Built-in Concurrency** – Managing **multiple connections, message ordering, and delivery** is manual and error-prone.

## Broker-Based Messaging Systems

Broker-based messaging systems rely on a **central message broker** to manage the communication between different services. The broker acts as an intermediary, receiving messages from producers and delivering them to consumers.

Common systems:

| Software | Protocol(s) Used |
|---|---|
| RabbitMQ | AMQP (Advanced Message Queuing Protocol), MQTT, STOMP |
| Apache Kafka/RedPanda | Kafka Protocol (custom TCP-based protocol) |
| ActiveMQ | AMQP, STOMP, MQTT, OpenWire |

Proprietary systems:

| Proprietary Managed Broker | Provider | Protocol(s) |
|---|---|---|
| Amazon SQS | AWS | HTTPS/REST API |
| Azure Service Bus | Microsoft | AMQP 1.0, HTTPS/REST API |
| Google Cloud Pub/Sub | Google | HTTPS/REST API, gRPC |

### Advantages

1. **Decoupling** – Producers and consumers interact only with the broker, **reducing spatial coupling** and simplifying maintenance and scaling.
2. **Reliability** – Brokers provide **message storage and delivery guarantees**, ensuring messages reach consumers even if they are temporarily unavailable.
3. **Scalability** – Brokers (e.g., Kafka partitions, RabbitMQ clusters) can **handle high throughput** and scale horizontally.

### Disadvantages

1. **Single Point of Failure** – Without replication/failover, the broker can **interrupt message flow** or become a bottleneck under heavy load.
2. **Latency Overhead** – Messages must pass through the broker, introducing **additional delay** in delivery.

**Example:**

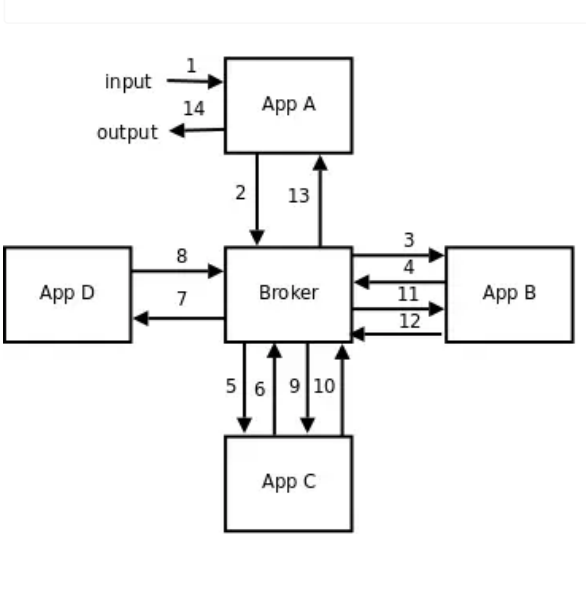
The following business logic implies many calls among services and the broker. Latency is added at each step. If the broker fails, the whole process fail.

```
function AppA (x) {
    y = do_business_logic_A (x);
    return AppB (y);
}

function AppB (x) {
    y = do_business_logic_B (x);
    return AppC (y);
}

function AppC (x) {
    y = do_business_logic_C (x);
    return AppD (y);
}

function AppD (x) {
    return do_business_logic_D (x);
}
```


input
1
14
output
App A
2
13
App D
8
7
Broker
3
4
11
12
App B
5
6
9
10
App C

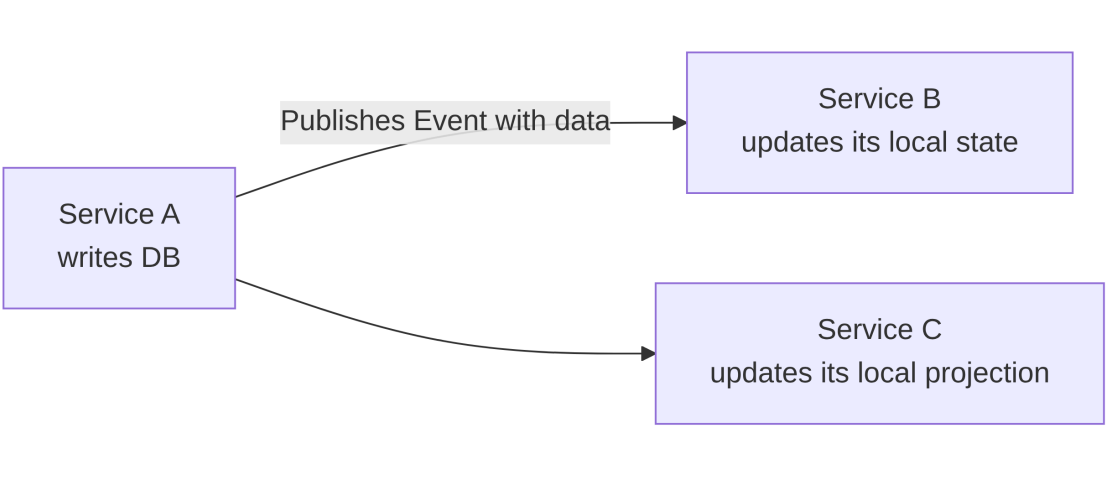
## Architectural Patterns

### Event-Carried State Transfer (ECST)

**Purpose:** Events *carry the state* needed by other services, so they maintain **local copies** and avoid synchronous calls.

**Characteristics:**

- Service A updates its state → publishes an event with the full relevant data.
- Service B consumes the event → updates its local copy.
- Events act as a source of truth


Service A writes DB
Publishes Event with data
Service B updates its local state
Service C updates its local projection

**State**

- Operational and authoritative data of a service
- Used by the service's business logic
- Acts as the source of truth for that service's responsibilities

**Projection**

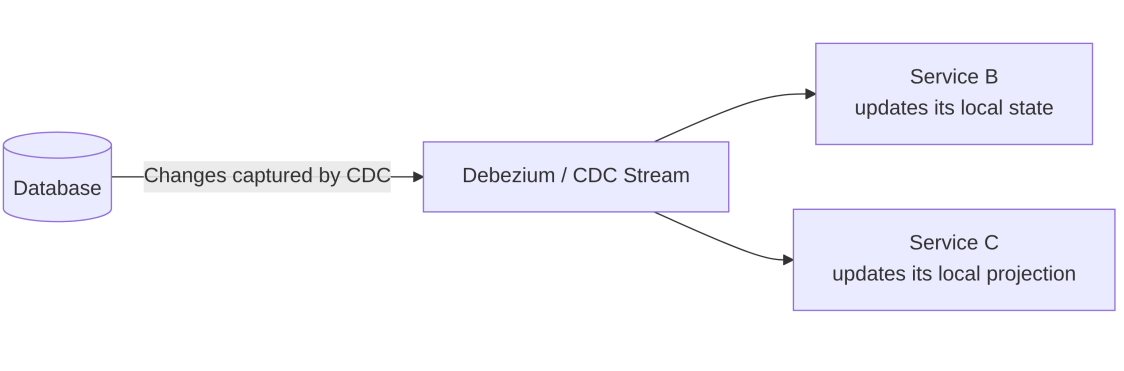
- Derived view of data built from events or other sources
- Optimized for querying, reporting, or read performance
- Not authoritative; can be rebuilt from the underlying events or state

### Change Data Capture (CDC) aka Automated ECST

**Purpose:**
Capture database changes (inserts, updates, deletes) and propagate them as events so other services or systems can **maintain synchronized copies** without direct synchronous access.

**Characteristics:**

- Database-centric
- Produces a stream of change events
- Supports near real-time replication
- Decouples consumers from the database

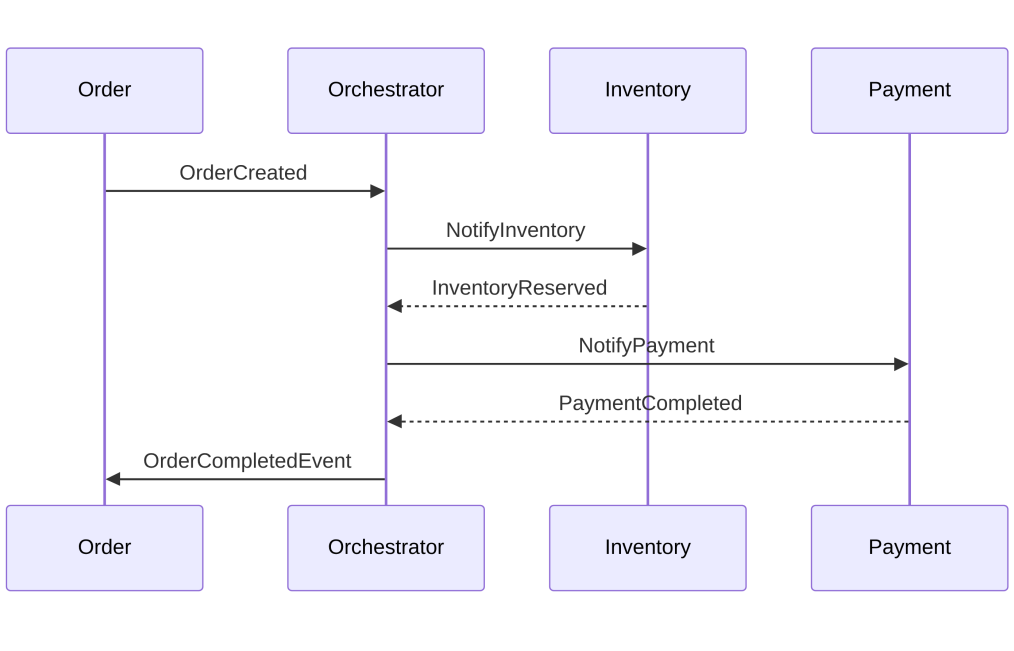

Database
Changes captured by CDC
Debezium / CDC Stream
Service B updates its local state
Service C updates its local projection

### Saga Pattern (Orchestration)

**Purpose:** A *central coordinator service* commands steps and compensations.

**Characteristics:**

- Workflow is easier to follow
- Logic centralized
- Better for long/complex processes

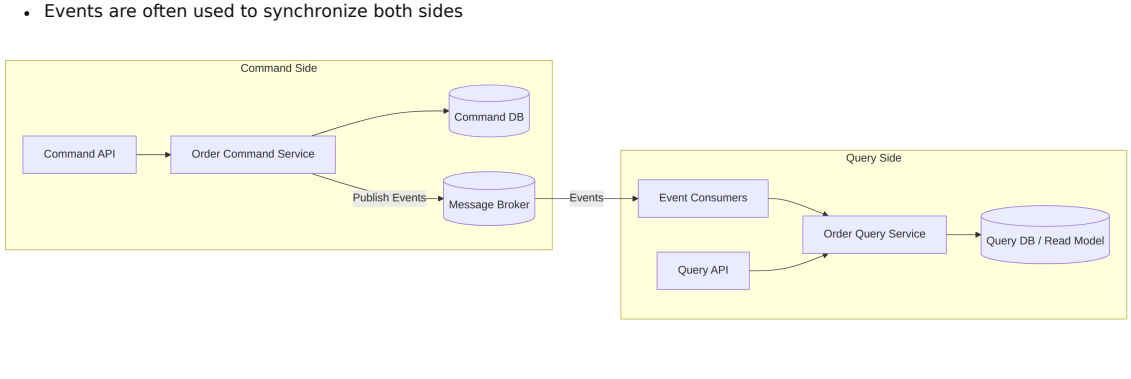

Order
Orchestrator
Inventory
Payment
OrderCreated
NotifyInventory
InventoryReserved
NotifyPayment
PaymentCompleted
OrderCompletedEvent
Order
Orchestrator
Inventory
Payment

### CQRS (Command Query Responsibility Segregation)

**Purpose:** Separate the system into a **write (command) side** and a **read (query) side** to optimize scalability, performance, and model clarity.

**Characteristics:**

- Commands and queries are handled by different models
- Write side focuses on consistency and business rules
- Read side is optimized for fast queries and projections
- Events are often used to synchronize both sides


Command Side
Command API
Order Command Service
Command DB
Publish Events
Message Broker
Events
Query Side
Event Consumers
Order Query Service
Query DB / Read Model
Query API

### Event Sourcing

**Purpose:** The *event log is the system of record*, not a table with current state.

**Characteristics:**

- Every state change = event
- State is rebuilt by event replay
- Perfect audit log
- Enables temporal queries

## Brokers at the Edge – Key Benefits

1. **Offline Resilience** – Buffer messages locally to survive **intermittent connectivity**.
2. **Resource Efficiency** – Queue messages asynchronously to avoid **blocking limited CPU/memory**.
3. **Bandwidth Optimization** – Batch or compress messages to **reduce network usage**.
4. **Local Security & Privacy** – Process sensitive data locally, sending only **encrypted updates** to the cloud.

## Resources